CPEN 416 / EEE 571O

Lecture 06: Multi-qubit measurements, the Bell basis, and superdense coding

Tuesday 23 September 2025

Announcements

- Midterm in class on Thursday 2 Oct (see PrairieLearn for details)
- Quiz 3 today
- Tutorial Assignment 1 due Friday 23:59

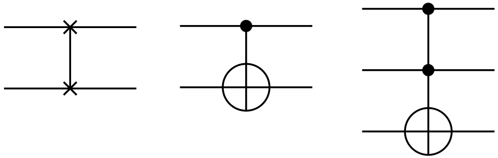
Last time

We used the tensor product to construct multi-qubit states.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\varphi\rangle = \gamma|0\rangle + \delta|1\rangle,$$

Last time

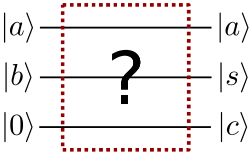
We saw examples of two-qubit gates:



Last time

X, CNOT, TOF can be used to create Boolean arithmetic circuits.

Exercise: use a combination of these gates to create a quantum version of a one-bit adder. Test it in PennyLane on all possible bit values of *a* and *b*.



Learning outcomes

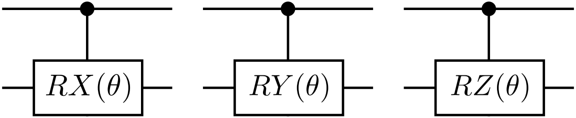
- apply gates controlled on the state of multiple qubits
- define and give examples of entangled states
- measure multi-qubit systems in the computational basis and Bell basis
- leverage entanglement to implement superdense coding

Example: controlled rotations (RX, RY, RZ)

We can control arbitrary Pauli rotations:

$$CRY(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ 0 & 0 & \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

Circuit elements:



PennyLane: `qml.CRX`, `qml.CRY`, `qml.CRZ`

Image credit: Codebook node I.14

Controlled-U

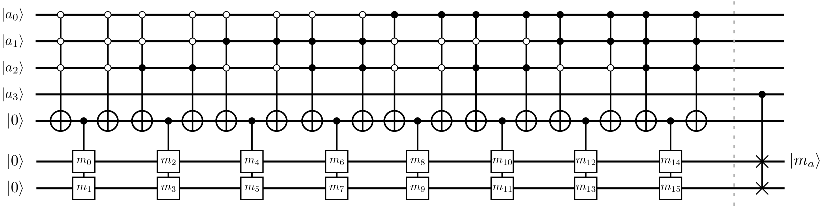
There is a pattern here:

$$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad CRY(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ 0 & 0 & \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

More generally,

Controlled unitary operations

Any unitary operation can be turned into a controlled operation, controlled on any state.



Most common controls are controlled-on-$|1\rangle$ (filled circle), and controlled-on-$|0\rangle$ (empty circle).

Controlled unitary operations with `qml.ctrl`

Remember `qml.adjoint`:

```
@qml.qnode(dev)
def my_circuit():
    qml.S(wires=0)
    qml.adjoint(qml.S)(wires=0)
    return qml.sample()
```

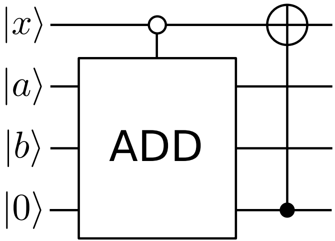
A similar *transform* allows us to perform arbitrary controlled operations (or entire quantum functions)!

```
@qml.qnode(dev)
def my_circuit():
    qml.S(wires=0)
    qml.ctrl(qml.S, control=1)(wires=0)
    return qml.sample()
```

Controlled unitary operations with `qml.ctrl`

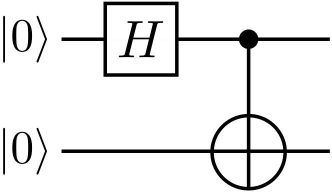
Take home exercise: using your adder from earlier, implement a circuit in PennyLane that uses `qml.ctrl` to

- perform addition if the control qubit is in state $|0\rangle$
- flip the state of the control qubit if the carry bit is 1



Controlling on superpositions

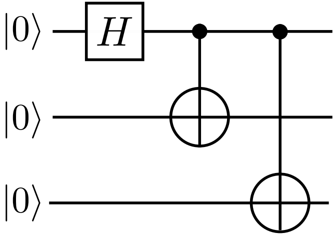
Exercise: What does CNOT do when the control qubit is in a superposition? Determine the combined output state of this circuit, then try to express it as a tensor product of two single-qubit states.



Entanglement

CNOT is an **entangling gate**!

Entanglement generalizes to more than two qubits:



Exercise: Express the output state of this circuit in the computational basis.

Multi-qubit measurement outcome probabilities

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

If we measure in the computational basis, the outcome probabilities are:

- $|\alpha|^2 = |\langle 00|\psi\rangle|^2$ for $|00\rangle$
- $|\beta|^2 = |\langle 01|\psi\rangle|^2$ for $|01\rangle$
- ...

Exercise: what is the probability of the qubits being in state $|110\rangle$ after measuring $(H \otimes SX \otimes I)|000\rangle$ in the computational basis?

Multi-qubit measurement outcome probabilities

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

We can also measure *just one qubit*:

- The probability of the first qubit being in state $|0\rangle$ is $|\alpha|^2 + |\beta|^2$
- The probability of the second qubit being in state $|1\rangle$ is $|\beta|^2 + |\delta|^2$

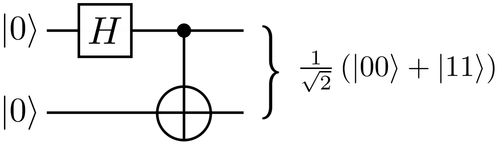
Exercise: what is the probability of the second qubit being in state $|0\rangle$ after measuring $(H \otimes SX \otimes I)|000\rangle$ in the computational basis?

Bell states

Remember how we created

$$|\Psi_{00}\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right),$$

from the $|00\rangle$ state:



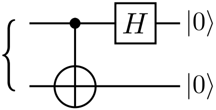
Bell states

Exercise: Apply the same circuit to the other 3 computational basis states? What is special about these four states?

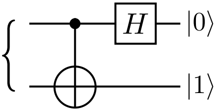
The Bell basis

We can measure in this basis by applying the adjoint of the circuit:

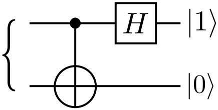
$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$



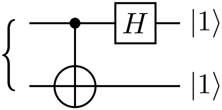
$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$



$\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$



$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$



Two famous quantum algorithms, **superdense coding** and **teleportation** work by performing a measurement in the Bell basis.

Superdense coding

Suppose Alice wants to send Bob two classical bits of information, say ‘1’ and ‘0’.

Q1: How many classical bits must she send to Bob to do this?

Q2: How many *qubits* must she send to Bob to do this?

Superdense coding

Alice and Bob start the protocol with this shared entangled state:

$$|\Phi\rangle_{AB} = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

Next, depending on her bits, Alice performs one of the following operations on her qubit:

$$\begin{aligned} 00 &\rightarrow I \\ 01 &\rightarrow X \\ 10 &\rightarrow Z \\ 11 &\rightarrow ZX \end{aligned}$$

Superdense coding

What happened to the entangled state?

$$|\Phi\rangle_{AB} = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

It will transform to:

$$00 \quad \rightarrow_I$$

$$01 \quad \rightarrow_X$$

$$10 \quad \rightarrow_Z$$

$$11 \quad \rightarrow_{ZX}$$

Superdense coding

Now, Bob can either

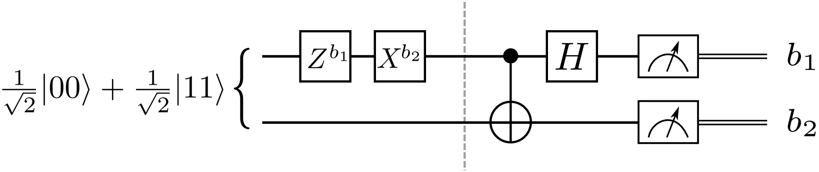
- perform a measurement directly in the Bell basis
- perform a basis transformation from the Bell basis back to the computational basis

to determine his state with certainty, and thus, the bits Alice sent.

$$
\begin{aligned}
(H \otimes I)\mathrm{CNOT} \cdot \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) &= |00\rangle \\
(H \otimes I)\mathrm{CNOT} \cdot \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle) &= |01\rangle \\
(H \otimes I)\mathrm{CNOT} \cdot \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) &= |10\rangle \\
(H \otimes I)\mathrm{CNOT} \cdot \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) &= |11\rangle
\end{aligned}
$$

Hands-on: superdense coding

Let's go implement it!



Next time

Content:

- Quantum teleportation

Action items:

1. Tutorial assignment 1 due Friday 23:59
2. Study for midterm

Recommended reading:

- Codebook nodes IQC, SQ, MQ; Nielsen & Chuang 1.2, 1.3.1-1.3.7, 2.2.3-2.2.5, 2.3, 4.3